MetaBreath AI Pipeline

Technical Research Report

Breath-Acetone Metabolic Risk Classification System

NSC 2026 — Cheewarun Health Platform
Version 1.0 | Report Date: 2026-07-12

Status: Production Demo — Models Trained & Deployed on FastAPI

Abstract

This report documents the complete AI pipeline of the MetaBreath device — a breath-acetone metabolic risk monitoring system targeting diabetes management. Four machine learning models are described: Random Forest (RF), XGBoost (XGB), a two-layer LSTM classifier, and a sensor drift detector. All models are trained, serialised, and deployed as REST API endpoints. This document explicitly addresses the distinction between trained production models and components requiring future pilot data, and presents full training provenance, hyperparameters, and quantitative performance metrics.

1. System Overview

The MetaBreath AI pipeline translates raw TGS1820 metal-oxide sensor readings from an ESP32 microcontroller into a clinically meaningful metabolic risk classification. The pipeline consists of four independent model components operating in a priority cascade, plus an LLM safety guardrail layer for the AI coaching interface.

1.1 Architecture Overview

Layer	Component	Training Status	Input	Output
1	Signal Processing Pipeline	N/A (deterministic)	Raw sensor voltage (V)	Normalised 13 features
2A	XGBoost Classifier	TRAINED ☐	13-feature snapshot	5-class label + confidence
2B	Random Forest Classifier	TRAINED ☐	13-feature snapshot	5-class label + confidence
3	LSTM Temporal Classifier	TRAINED ☐*	Sequence of 5 readings × 8 features	3-class ☐ refined 5-class
4	Drift Detector	TRAINED ☐	Calibration history (ambient VOC)	Drift %, severity, recommendation
5	Anderson Rule-Based (fallback)	Deterministic	Acetone delta (ppm)	5-class label
6	LLM Safety Guardrail	Regex + prompt eng.	User / LLM message text	Block/pass + disclaimer

*LSTM trained on surrogate lab data; real-world pilot data collection planned post-NSC.

2. Training Data Provenance

Important Disclosure (Addressing Reviewer Feedback)
The current training dataset does NOT contain breath samples measured with the MetaBreath device itself. Data sources are: (1) a synthetic demo dataset generated from Anderson 2015 clinical thresholds (primary RF/XGB training), (2) a Kaggle eNose dataset containing real breath measurements from 1,000 human subjects (545 diabetic, 455 normal) using TGS-family sensors — referenced for TGS family response alignment, and (3) UCI Gas Drift data for drift detector training. A controlled pilot study with MetaBreath hardware and real participants is the planned next phase. This limitation is explicitly stated to enable honest feasibility assessment.

2.1 Dataset Summary

Dataset	Source	Rows	License	Used For	Type
MetaBreath Demo	NSC 2026 official synthetic dataset	1,199	NSC Internal	RF / XGB primary training	Synthetic

eNose Diseases (Kaggle)	Muhammad Rizwan (Apache-2.0)	1,000	Apache-2.0	Reference — TGS family response alignment (not merged in current pipeline)	Human breath (545 diabetic / 455 normal)
UCI Gas Drift	UCI ML Repo / Vergara (CC BY 4.0)	13,910	CC BY 4.0	Drift Detector training (Acetone gas type 5)	Lab gas
Total		16,109			

2.2 Label System — Anderson 2015 Five-Class

All models converge on the Anderson 2015 five-pattern breath-acetone classification (doi:10.1002/oby.21242). The threshold system provides a deterministic ground-truth label from any acetone-ppm measurement, enabling unified label mapping across heterogeneous datasets.

Class	Acetone Range (ppm)	Clinical Meaning	Integer Index
basal	0.5 – 2.0	Standard diet, basal ketosis	0
light_ketosis	2.0 – 4.0	Mild caloric restriction	1
nutritional_ketosis	4.0 – 30.0	HFLC/keto diet, BOHB 0.5–3 mM	2
deep_ketosis	30.0 – 75.0	Fasting / extended restriction	3
dka_risk	≥ 75.0	DKA range — medical attention required	4

Source: Anderson JC. Obesity (2015) 23:2327–2334.

2.3 Baseline Construction

The sensor baseline (ambient VOC) is established at device boot via a 10-second clean-air calibration cycle on the TGS1820 sensor. The acetone_delta feature is computed as:

```
acetone_delta = (sensor_voltage − baseline_voltage) × gain + offset
```

where gain and offset are per-device calibration coefficients stored in flash. Temperature and humidity compensation from SHT31 sensor is applied as:

```
VOC_comp = VOC_raw / [(1 + 0.015·ΔT) × (1 + 0.008·ΔH)]
```

where $\Delta T = T - 20°C$ and $\Delta H = H - 65\%RH$ (TGS1820 datasheet coefficients). In the training datasets, baseline is treated as the minimum VOC reading over the first 3 calibration epochs per session.

2.4 Train/Test Split

Model	Total Rows	Train	Test	Split Method	Stratified
RF / XGB	2,199	1,759 (80%)	440 (20%)	train_test_split, seed=42	Yes
LSTM Classifier	2,199*	~1,870 (85%)	~330 (15%)	train_test_split, seed=42	Yes

Drift Detector	3,009 (acetone rows)	~2,407 (80%)	~602 (20%)	batch-aware split	N/A

*LSTM uses same merged dataset; sequences are constructed with a 5-reading sliding window.

3. Model Specifications

3.1 Random Forest Classifier

Hyperparameter	Value	Rationale
n_estimators	200	Balance bias-variance; avoids overfitting small dataset
max_depth	8	Constrain tree depth to prevent memorisation
min_samples_leaf	5	Minimum 5 samples per leaf for generalisation
class_weight	'balanced'	Compensate class imbalance across 5-class output
random_state	42	Reproducibility
n_jobs	-1	All CPU cores for parallel tree training

3.2 XGBoost Classifier (Optuna-tuned)

XGBoost hyperparameters were found via Optuna Bayesian optimisation over 50 trials (5-fold CV F1-weighted objective, timeout 300s).

Hyperparameter	Search Space	Best Value Found
n_estimators	100 – 500	Optuna-selected
max_depth	3 – 10	Optuna-selected
learning_rate	0.01 – 0.30 (log)	Optuna-selected
subsample	0.5 – 1.0	Optuna-selected
colsample_bytree	0.5 – 1.0	Optuna-selected
reg_alpha	1e-4 – 10.0 (log)	Optuna-selected
reg_lambda	1e-4 – 10.0 (log)	Optuna-selected
scale_pos_weight	class freq. ratio	Auto from training labels
eval_metric	—	logloss
random_state	—	42
Optuna trials	—	50 (timeout 300 s)
CV strategy	—	5-fold, stratified, F1-weighted

3.3 LSTM Temporal Classifier

LSTM Training Status: TRAINED (surrogate data) — Pilot Data Required

The LSTM model file (lstm_model.pt) is trained and deployed in production. However, training used lab-controlled surrogate data, not breath samples from real patients. The model classifies 5-reading sequences correctly for stable low/high patterns (F1=0.9722, val_acc=0.9565) but shows weakness in detecting ramping/transitional sequences — a known limitation from the absence of real-world transition-pattern training examples.

3.3.1 Architecture (PyTorch)

Layer	Type	Configuration
Input	—	Shape (batch, 5, 8) — 5 readings × 8 features
Layer 1	LSTM	input_size=8, hidden_size=64, batch_first=True
Dropout 1	Dropout	p=0.30
Layer 2	LSTM	input_size=64, hidden_size=32, batch_first=True
Dropout 2	Dropout	p=0.30 (applied to last time-step only)
FC 1	Linear	in=32, out=16
Activation	ReLU	—
FC 2 (output)	Linear	in=16, out=3 (low / moderate / high)
Total parameters	—	~37,000 trainable parameters

3.3.2 Training Configuration

Parameter	Value	Notes
Loss function	CrossEntropyLoss	3-class: low / moderate / high
Optimiser	Adam	lr=1e-3, default β□=0.9, β□=0.999
Epochs	100 (max)	EarlyStopping: patience=10, restore_best=True
Batch size	64	—
LR scheduler	ReduceLROnPlateau	factor=0.5, patience=5
Validation split	15%	Random split from training set, seed=42
Input features (8)	acetone_delta, quality_score, reliability_score, ketosis_index, metabolic_score, pressure_mean, temperature, humidity	
Preprocessing	StandardScaler	Fit on training split; saved as scaler_lstm_mean.npy + scale.npy
Sequence length	5 readings	Minimum 5 to invoke LSTM; <5 falls back to XGBoost
Class mapping	0=low, 1=moderate, 2=high	Refined post-hoc to 5-class via Anderson threshold
Framework	PyTorch	Model saved as lstm_model.pt

3.3.3 LSTM Feature Details

Feature	Source	Unit	Description
acetone_delta	TGS1820 + calibration	ppm	Primary metabolic signal; baseline-subtracted VOC
quality_score	Signal processing	0–100	Reading quality from voltage, pressure, env. conditions

reliability_score	Signal processing	0–100	Combines quality, drift, and calibration age
ketosis_index	Derived	0–1	Normalised ketosis proximity score
metabolic_score	Derived	0–100	Composite metabolic activity score
pressure_mean	XGZP6847A	kPa	Mean breath differential pressure during measurement
temperature	SHT31	°C	Ambient temperature for compensation
humidity	SHT31	%RH	Ambient humidity for compensation

3.4 Sensor Drift Detector

The drift detector quantifies sensor calibration degradation over time. It operates in parallel with the risk classifier and provides recalibration recommendations. The primary mechanism compares recent ambient VOC readings against a 3-reading baseline established at first calibration.

Drift %	Severity	Recommendation	Training Data Source
< 10%	none	ok	UCI Gas Drift — Acetone (batch 1–10)
10% – 25%	mild	recalibrate_soon	UCI Gas Drift — batch 7–8 degradation pattern
> 25%	severe	recalibrate_now	UCI Gas Drift — batch 9–10 degradation pattern

3.5 LSTM Trend Classifier (Phase 3 Redesign)

Phase 3 Redesign — LSTM as Trend Classifier
The legacy per-reading LSTM (§3.3) duplicated the RF/XGB Anderson rule and shared the same label-feature circularity (L9). Phase 3 reframes the LSTM's task to what a temporal model is uniquely suited for: classifying the direction of the user's own baseline over time, not a per-reading class. Output is 4-class softmax [stable, increasing, decreasing, abnormal]. Labels are derived from an OLS slope + spike-vs-median rule applied to the whole sequence (app/services/trend_label.py), so no single input feature is a deterministic function of the label — breaking the circularity that still applies to §3.3.

3.5.1 Architecture (PyTorch)

Layer	Type	Configuration
Input	—	Shape (batch, L, 8) — L □ {7, 14, 30}
Layer 1	LSTM	input_size=8, hidden_size=64, batch_first=True
Dropout 1	Dropout	p=0.30
Layer 2	LSTM	input_size=64, hidden_size=32, batch_first=True
Dropout 2	Dropout	p=0.30 (last time-step only)
FC 1	Linear + ReLU	in=32, out=16
FC 2 (output)	Linear	in=16, out=4 (stable / increasing / decreasing / abnormal)
Total parameters	—	~38,000 trainable parameters

3.5.2 Training Configuration

Parameter	Value	Notes
Loss function	CrossEntropyLoss	4-class trend
Optimiser	Adam	lr=1e-3, β□=0.9, β□=0.999
LR scheduler	ReduceLROnPlateau	factor=0.5, patience=5
Batch size	16	—
Epochs	150 max	EarlyStopping patience=15 on val_loss
Input features (8)	ΔVOC, pressure_mean, pressure_std, breath_duration, temperature, humidity, quality_score, reliability_score	
Preprocessing	StandardScaler	fit on TRAIN patients only
Sequence length	14 sessions (min 7)	runtime accepts 7–30; <7 □ insufficient_data
Label rule	OLS slope + max-jump / median-jump ratio	slope>0.30 □ increasing; slope<−0.30 □ decreasing; abs jump>4 ppm and >3× median □ abnormal; else stable
Framework	PyTorch	saved as lstm_trend.pt

3.5.3 Validation Strategy — Participant-wise Split

Training patients and validation patients are DISJOINT sets — 80/20 stratified split by patient_id. This differs from the random within-patient split used by §3.3 and prevents the model from memorising per-person quirks and reporting them as generalisation. Synthetic dataset: 100 virtual patients × 14 sessions/patient = 1,400 rows (balanced 25/25/25/25 across trend classes). See plan.md §5.3 for the generation rules and tests/test_lstm_trend_inference.py for canonical-pattern assertions.

4. Model Performance Metrics

The following metrics are from training runs on the merged surrogate dataset. Metrics represent model performance on the held-out test split (20% / 15% depending on model). Cross-validation (CV) scores reflect 5-fold stratified CV on the training split only.

> Interpretation Note — Rule-Verification vs. Predictive Validity
> Training labels are derived by applying the Anderson 2015 threshold rule directly to acetone_delta, which is itself one of the model input features. Two RF/XGB variants are therefore trained and reported side-by-side: (a) a verification variant that keeps all 13 features and measures how reliably the classifier reproduces the Anderson rule under sensor noise; and (b) a predictive variant in which acetone_delta and its three derived features (ketosis_index, metabolic_score, fat_burning_index) are removed, so no feature is a function of the label. On the current synthetic dataset the verification variant scores 0.99 and the predictive variant collapses to 0.40 — essentially the stratified chance baseline of 0.3783 — which precisely quantifies the label-feature circularity. Independent predictive validity can only be established after retraining with a label that is not a function of acetone_delta, such as blood BOHB / GC-MS collected during the pilot phase (Section 7, L9).

4.1 Summary Performance Table

Model	Variant	Test Accuracy	F1-weighted (test)	CV F1 Mean ± SD	n_train	n_test
Random Forest	verification (13 feat.)	0.9917	0.9917	0.9907 ± 0.0063	959	240
XGBoost	verification (13 feat.)	0.9917	0.9903	0.9926 ± 0.0061	959	240
Random Forest	predictive (9 feat.)	0.3958	0.3969	0.3725 ± 0.0456	959	240
XGBoost	predictive (9 feat.)	0.4333	0.3737	0.3848 ± 0.0043	959	240
Chance baseline	stratified 5-class	0.3783	—	—	—	240
LSTM (legacy)	per-reading 3-class	0.9722	0.9722	val_acc = 0.9565 *	~1,870	~330
LSTM Trend	4-class trend (Phase 3)	0.9500	0.9495	val (participant) *	80 pt	20 pt
Drift Detector	drift	0.9850	0.9850	0.8418 *	~2,407	~602

Reading this table. Verification-variant metrics quantify how reliably the deployed classifier reproduces the Anderson threshold rule given noisy inputs; they are not evidence of independent predictive validity (see Interpretation Note above and L9 in Section 7.1). Predictive-variant metrics remove acetone_delta, ketosis_index, metabolic_score, and fat_burning_index — the four features that are either the label source or deterministic functions of it — and measure whether the remaining sensor/environment features carry any independent signal for Anderson classes. Predictive-variant accuracy (0.40–0.43) sits at the stratified chance baseline (0.3783), which is the expected outcome on the current synthetic dataset and quantifies exactly how much of the verification-variant score is attributable to circularity. * LSTM CV = validation-set accuracy (15% split), not 5-fold CV. Drift detector CV = held-out batch performance.

4.2 LSTM Temporal Test Scenarios (System-Level)

In addition to model-level metrics, the full pipeline was tested through 15 simulation scenarios covering clinical edge cases:

Scenario	Input Sequence	Expected	Got	Confidence	Result
Legacy — stable healthy (5 days)	0.5–0.9 ppm	low	low	0.999	PASS
Legacy — ramping into ketosis	2□40 ppm	not low	low	0.630	FAIL †
Legacy — consistently high risk	80–100 ppm	high	high	1.000	PASS
Legacy — short sequence (2)	<5 readings	fallback	xgb_fallback	—	PASS
Trend — stable 14 days	0.5–2 ppm × 14	stable	stable	0.952	PASS
Trend — ramping 2□41 ppm	2 + 3·t × 14	increasing	increasing	0.950	PASS ‡
Trend — decreasing 28□2 ppm	28 − 2·t × 14	decreasing	decreasing	0.709	PASS
Trend — spike at day 7 (+15)	stable + 1 spike	abnormal	abnormal	0.885	PASS
Trend — short sequence (5)	<7 sessions	insufficient_data	insufficient_data	—	PASS

† FAIL preserved for transparency: the legacy per-reading LSTM classified a ramping 2□40 ppm sequence as 'low' because its training data lacked gradual-transition examples. ‡ RESOLVED in Phase 3: the new Trend LSTM (Section 3.5) reframes the LSTM's task from per-reading classification to 4-class trend detection over a 7–30 session window; the same ramp is now correctly labelled 'increasing' at confidence 0.95. Limitation L4 (Section 7.1) is therefore addressed.

4.3 Full Pipeline Simulation (5-Day Patient)

Day	Acetone (ppm)	Anderson Label	XGBoost	LSTM	Confidence	Match?
0	0.6	basal	low	low	1.000	YES
1	2.5	light_ketosis	low	low	1.000	YES
2	8.0	nutritional_ketosis	low	low	1.000	YES
3	25.0	nutritional_ketosis	low	low	1.000	YES
4	55.0	deep_ketosis	moderate	moderate	1.000	YES

XGBoost and LSTM predictions matched 5/5 days. Note: 3-class LSTM output maps to different labels than 5-class Anderson — both 'low' values here represent labels in the 0–30 ppm Anderson range.

4.4 Drift Detection Performance

Scenario	Drift %	Expected Severity	Got	Result
Stable sensor	0.23%	none	none	PASS
Mild drift (+15%)	15.12%	mild	mild	PASS

Severe drift (+40%)	44.19%	severe	severe	PASS
Insufficient data (1)	—	insufficient_data	insufficient_data	PASS

5. Feature Engineering (13 Features for RF/XGB)

Features marked ● in the last column are excluded from the predictive variant (§4.1). These four are either the label source (acetone_delta) or deterministic functions of it (ketosis_index, metabolic_score, fat_burning_index); keeping them creates label–feature circularity (see §7.1 L9).

#	Feature Name	Sensor / Origin	Unit	Description	Missing	Leaky
1	acetone_delta	TGS1820	ppm	Primary signal: baseline-subtracted VOC, env-compensated	Default 0	●
2	quality_score	Signal processing	0–100	Composite reading quality (voltage, pressure, env.)	Default 100	
3	reliability_score	Signal processing	0–100	Quality + drift penalty + calibration age	Default 100	
4	ambient_voc	TGS1820 (clean air)	ppm	Baseline ambient VOC (calibrated pre-measurement)	Default 0	
5	pressure_mean	XGZP6847A	kPa	Mean breath differential pressure	Default 0	
6	pressure_std	XGZP6847A	kPa	Std dev of breath pressure (effort consistency)	Default 0	
7	breath_duration	Firmware timer	s	Duration of breath measurement cycle	Default 3	
8	temperature	SHT31	°C	Ambient temperature	Default 20	
9	humidity	SHT31	%RH	Ambient relative humidity	Default 65	
10	environment_penalty	Derived	0–50	Distance from ideal env. (20°C, 65%RH)	Computed	
11	ketosis_index	Derived	0–1	Normalised proximity to nutritional ketosis zone	Default 0	●
12	metabolic_score	Derived	0–100	Composite from acetone + quality + context	Default 0	●
13	fat_burning_index	Derived	0–1	Estimated fat-oxidation intensity from acetone delta	Default 0	●

6. Inference Priority Cascade

The system uses a priority cascade to maximise robustness. At each tier, if the model is unavailable or returns low confidence, the system falls through to the next tier.

Priority	Model	Trigger Condition	Fallback Trigger
0 (gate)	Reliability Gate	Always first — blocks low-quality readings	reliability_score < 40 ☐ return 'unreliable'
1	XGBoost	Model loaded + reliability ≥ 40	model None or exception ☐ try RF
2	Random Forest	XGBoost unavailable	model None or exception ☐ rule-based
3 (LSTM)	LSTM (temporal)	≥5 readings available via /ai/predict/lstm	< 5 readings or model None ☐ XGBoost_fallback
4	Anderson Rule-Based	All ML models failed/missing	Final deterministic fallback — always returns

Confidence threshold: any model returning confidence < 0.60 produces label = 'unreliable' and sets recalibration_needed = True. The 0.60 threshold was chosen to maintain precision at the cost of recall on ambiguous readings, consistent with clinical conservatism.

6.1 API Endpoints

Endpoint	Method	Model	Input	Response Fields
/ai/predict	POST	XGBoost ☐ RF ☐ rule-based	Single reading (13 features)	label, metabolic_risk_index, confidence_score, model_used
/ai/predict/lstm	POST	LSTM ☐ XGBoost_fallback	Sequence of ≥5 readings	label, metabolic_risk_index, confidence_score, sequence_length
/ai/predict/trend	POST	LSTM Trend ☐ rule fallback	Sequence of ≥7 sessions	trend, confidence, probabilities, sequence_length
/ai/trend	GET	Linear regression	≥3 historical readings	trend_direction, slope_ppm_per_day, predicted_points, confidence
/ai/drift	GET	Heuristic + XGBoost drift model	Calibration history	drift_detected, severity, drift_pct, recommendation
/ai/chat	POST	LLM + guardrail	User message + sensor data	ai_response (filtered), disclaimer

7. Limitations and Future Work

7.1 Current Limitations

#	Limitation	Impact	Mitigation
L1	No real human breath data	High — training on surrogate data; clinical validity unproven	Planned pilot study with human participants
L2	No real participant count	High — cannot claim clinical sensitivity/specificity	Pilot: target 30+ subjects with confirmed metabolic states
L3	Threshold-based ground truth	Medium — labels derived from Anderson thresholds, not GC-MS verification	Require blood ketone / GC-MS co-measurement in pilot
L4	LSTM ramp detection weak (legacy §3.3)	ADDRESSED in Phase 3 — the new Trend LSTM (§3.5) reframes the task as 4-class trend classification over 7–30 sessions; the 2□40 ppm ramp is now correctly labelled 'increasing' at confidence 0.95 (§4.2). Legacy per-reading LSTM row retained for continuity.	Real longitudinal per-participant data from pilot phase will replace the synthetic dataset used to train the current Trend LSTM (§7.2).
L5	No DKA range data (>75 ppm)	High — no training examples for highest-risk class	Cannot resolve without clinical data; flag as unvalidated range
L6	Drift model = heuristic fallback	Low — XGBoost drift model trained but feature mismatch with real sensor	Retrain drift model after pilot with real MetaBreath calibration logs
L7	No Confusion Matrix published	Medium — reviewer cannot fully assess per-class errors	See Section 4 for F1-scores; confusion matrix plots in model output files
L8	No ROC-AUC for 5-class	Low — standard ROC-AUC requires binary or OvR	Report macro-averaged OvR AUC in next version after pilot
L9	Label–feature circularity (RF / XGB)	High — labels are computed by applying the Anderson threshold to acetone_delta, which is also a model input feature. Verification-variant accuracy (RF=0.9917, XGB=0.9917) therefore measures rule-consistency, not independent predictive validity. The predictive variant (RF=0.3958, XGB=0.4333) sits at the stratified chance baseline (0.3783), quantifying the drop.	Predictive variant already trained and reported in §4.1 as an honest baseline. Independent predictive validity requires retraining with a non-circular label (blood BOHB / GC-MS) collected during the pilot phase (§7.2).

7.2 Pilot Study Plan (Next Phase)

Item	Details
Target participants	≥30 volunteers; mix of healthy, keto-diet, and Type-2 diabetic (supervised)
Sessions per participant	≥5 breath measurements at different metabolic states (fasting, post-meal, post-exercise)
Reference standard	Concurrent blood ketone (Precision Xtra) + urine strip (Ketostix) per session
Sensor validation	Compare MetaBreath acetone_delta against laboratory GC-MS reference (subset)
Labels	Ground-truth from blood BOHB: <0.5 mM=basal, 0.5–3 mM=nutritional_ketosis, >3 mM=deep
LSTM re-training	Collect 5+ consecutive daily readings per participant for temporal sequence training
Ethics	IRB approval required; no clinical diagnosis made from device during pilot
Target timeline	Post-NSC 2026; Phase 6 of Cheewarun development roadmap

8. LLM Safety Guardrail — Cheewarun AI Coach

The Cheewarun AI Coach uses an LLM (Claude) for natural-language interpretation of sensor readings. A mandatory safety guardrail layer pre-screens both user input and LLM output for medically dangerous content.

Category	Examples Blocked (EN)	Examples Blocked (TH)
Drug dosage	"adjust insulin dose", "how much metformin"	"ปรับยา", "ฉีดอินซูลินเท่าไหร่"
Diagnosis	"you have diabetes", "DKA"	"คุณเป็นเบาหวาน", "เป็น DKA"
Deny medical need	"don't need a doctor"	"ไม่ต้องไปหาหมอ"
Extreme fasting	"fast for 7 days"	"อดอาหาร 7 วัน"
Self-harm	"kill myself", "self-harm"	"อยากตาย", "ฆ่าตัวตาย"

All responses include a mandatory medical disclaimer in Thai and English. Emergency symptoms trigger immediate referral: "โปรดโทร 1669 หรือไปห้องฉุกเฉินทันที".

9. Model File Inventory

File	Size	Contents	Status
apps/api/models/rf_classifier.joblib	~1.2 MB	RF verification variant (13 feat.) — deployed	Production
apps/api/models/xgb_classifier.joblib	~0.8 MB	XGB verification variant (13 feat.) — deployed	Production
apps/api/models/rf_classifier_predictive.joblib	~1.0 MB	RF predictive variant (9 feat., leakage-free) — reporting only	Baseline
apps/api/models/xgb_classifier_predictive.joblib	~0.7 MB	XGB predictive variant (9 feat., leakage-free) — reporting only	Baseline
apps/api/models/lstm_model.pt	~0.5 MB	Legacy per-reading LSTM (3-class metabolic)	Production*
apps/api/models/lstm_trend.pt	~0.6 MB	Trend LSTM (4-class trend, Phase 3, participant-wise validated)	Production
apps/api/models/lstm_trend_metrics.json	<1 KB	Trend LSTM val metrics + confusion matrix	Reference
data/processed/scaler_lstm_trend_mean.npy	<1 KB	Trend LSTM StandardScaler mean	Required
data/processed/scaler_lstm_trend_scale.npy	<1 KB	Trend LSTM StandardScaler scale	Required
data/processed/longitudinal_synthetic.csv	~100 KB	Synthetic longitudinal dataset (100 pt × 14 sess)	Reproducible
apps/api/models/drift_model.joblib	~0.3 MB	Trained drift detector (XGBoost)	Production
apps/api/models/feature_columns.json	<1 KB	Feature order + label encoder classes	Required
apps/api/models/training_metrics.json	<1 KB	RF/XGB training metrics snapshot	Reference
data/processed/scaler_lstm_mean.npy	<1 KB	LSTM StandardScaler mean	Required
data/processed/scaler_lstm_scale.npy	<1 KB	LSTM StandardScaler scale	Required
apps/api/notebooks/01_prepare_data.ipynb	—	Data merging + feature engineering	Reproducible
apps/api/notebooks/02_random_forest.ipynb	—	RF training + evaluation	Reproducible
apps/api/notebooks/03_xgboost_optuna.ipynb	—	XGBoost + Optuna tuning	Reproducible
apps/api/notebooks/04_lstm_temporal.ipynb	—	LSTM training + evaluation	Reproducible

* LSTM in production but trained on surrogate data. Refresh required after pilot study.

10. References

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